

Continuous improvement through Lean Six Sigma: a systematic literature review and bibliometric analysis

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ABSTRACT (ENGLISH)

Purpose

Lean Six Sigma (LSS) improves the efficiency of organizations by systematically reducing variability, optimizing processes and eliminating waste to promote quality, productivity and customer satisfaction. The purpose of this paper is to investigate the impacts of LSS on promoting continuous improvement practices and enhancing organizational efficiency.

Design/methodology/approach

The method of this study involves a comprehensive systematic literature review and bibliometric analysis, which encompasses the time period between 2001 and April 2024 and collecting 1,992 articles from the Scopus database. The study uses several advanced tools such as RStudio for data processing, Biblioshiny for interactive analysis of bibliography and VOSviewer for illustrating scientific landscapes and research clusters detection.

Findings

The findings of the study point out that efficiency, quality improvement and cost minimization are significantly affected through the application of LSS tools and techniques, which is consequently leads to promoting organizational excellence through continuous improvement across a variety of industrial sectors. Furthermore, the study focuses on the recent trends in the LSS's application including the integration of digital tools with LSS to grow the significance on sustainability. These trends are following to find out directions for further research by fostering continuous development in the business processes.

Originality/value

The study is distinct in terms of its originality as systematic literature review and bibliometric analysis are coordinately demonstrated to provide a robust understanding about the impacts of LSS implementation in promoting organizational efficiency. The findings of the study generate new opportunities for the LSS sphere in the coming years. Using of modern technologies including Biblioshiny, VOSviewer and RStudio to analyze nearly two decades' scientific literature. Thus, valuable insights offered by this research into the practical implications of LSS tools from the economic and social context to boost organizational performance through continuous improvement.

FULL TEXT

1. Introduction and background of the study

Since the 1980s, the popular and famous Japanese term Lean Six Sigma (LSS) has undergone momentum as a powerful method in the discourse of academics and practitioners in the field of business management (Laureani and Antony, 2019 ; Madhani, 2022 ; Niñerola *et al.*, 2021b ; Tennant, 2017 ; Thomas *et al.*, 2016). LSS is also considered a valuable aspect of total quality management (TQM) that focuses on continuous development and cost-saving through offering quality and reliable products to satisfy the stakeholders (Alblooshi *et al.*, 2021 ; Raval and Kant, 2017). It is a meticulously structured

and disciplined approach which enhances the efficiency and effectiveness by improving business processes, minimizing waste in operations and identifying and removing the defects from the organizations (Jirasukprasert *et al.*, 2014 ; Prasad *et al.*, 2020 ; Ruben *et al.*, 2018). In addition, by reducing the defects level, LSS helps the business processes to provide high-quality products in the competitive markets (De Mast *et al.*, 2022 ; Hernandez-Lara *et al.*, 2021). With this method, organizations may precisely meet the customer expectations, adjust to shifting marketing conditions and enhance product quality. Therefore, LSS ensures long-term sustainability and competitiveness by continuously refining the processes and aligning them with business goals (De Koeijer *et al.*, 2014 ; Sharma and Gupta, 2015 ; Verma *et al.*, 2022), enabling organizations to succeed in dynamic circumstances (Escobar *et al.*, 2022 ; Meier *et al.*, 2023). For example, Motorola and General Electric have played a crucial role in the development and versatile application of the LSS method through achieving continuous improvement, which is named into DMAIC (Define, Measure, Analyze, Improve and Control) process (Hernandez-Lara *et al.*, 2021 ; Pongboonchai-Empl *et al.*, 2023). Manufacturing organizations widely adopt and implement the DMAIC approach to minimize the process variations (Purba *et al.*, 2021). Thus, due to adaptability and effectiveness of LSS, it has become a handy tool (Prabakar and Prabakar, 2023) to the current competitive organizations (Escobar *et al.*, 2022 ; Meier *et al.*, 2023).

Fostering a culture of ongoing development in business organizations, importance of LSS tools and techniques must be required to adapt the conditions of challenging environments and pursue the sustainable growth (Albliwi *et al.*, 2015 ; Rifqi *et al.*, 2024). In the era of fourth industrial revolution, LSS not only focuses on the traditional roots in manufacturing but also emphasizes on health care, manufacturing, finance, telecommunications and services, engineering and educational institutions (Antony, 2014 ; Antony *et al.*, 2012 ; Malik *et al.*, 2024 ; Prabakar and Prabakar, 2023 ; Sunder and Antony, 2020). The application of LSS in these diverse areas reveals the current growing trends for further comprehensive investigation on the tools and techniques of it (Raval *et al.*, 2018). However, researchers are more eager to explore the integrated outcome by aligning LSS approach with digital technologies such as artificial intelligence, data analytics and automation for enhancing process efficiency, improving decision-making with driving innovation (Fayyaz *et al.*, 2024 ; Skalli *et al.*, 2024). Therefore, the growing body of research shows the relevance and adaptability of LSS in modern organizations, wherein continuous improvement remains still as a primary position beyond of everything.

According to Parmar and Desai (2020) , the 18-year review of LSS studies highlights the limited integration of lean, six sigma and sustainability, while indicating the opportunities for further investigation using quantitative methods, blueprint modeling and implementing sustainable LSS models, in the automobile and construction sectors. Moreover, from the previous studies, it is found that adoption of LSS tools and techniques in human resources (HR), education, finance, manufacturing and health care to identify the key barriers, benefits and a research gap for further research to explore the literature and implementation of LSS (Marolla *et al.*, 2022 ; Singh and Rathi, 2019). Another bibliometric Analysis was reviewed which conducted by scholars (Singh and Ravi, 2023), wherein applications of LSS in hospitals during the past decade (2011–2020), emphasizing on list of author, author productivity, keyword clusters, publication trends for providing the opportunities for future research areas. In addition, it was analyzed from existing literature based on philosophies of some quality practitioners, they have created a difference between two approaches such as Six Sigma (SS) and Total Quality Management (TQM) by providing some misconceptions for quality improvement in the service industry (Antony, 2006). But, the researchers Albliwi *et al.* (2015) studied to find out key common themes including motivation factors, opportunities, barriers and forthcoming factors under the manufacturing areas for the LSS implementation through reflecting the research gaps for further study as holistic business improvement technique.

Based on review of existing literature on the LSS, it is still not uncovered that the impact of continuous improvement comprehensively through the implementation of tools and techniques of LSS in the business organizations. So, this indicates a research gap for further exploration and the current study as well. Then, this research paper is designed by concentrating on this research gap to show the extensive impact of continuous improvement through LSS following a systematic review and bibliometric analysis spanning from 2001 to April 2024. Based on the research gap, the research question is:

RQ1. What is the impact of continuous improvement through LSS in business organizations?

So, based on research question, the main objective of this study is to show impact comprehensively of continuous

improvement through LSS methodology in business organizations.

Although numerous previous studies have been conducted that analyze the implementation of the lean six-sigma approach in the business field those are not completely conclusive. In contrast to earlier studies, this study will provide a thorough overview about the successful implementation of LSS to influence continuous improvement by combining systematic review and bibliometric analysis. From the economic perspective, a new contribution will be created by examining the impact of LSS drives continuous improvement leading to gain efficiency and cost reductions in waste management within business processes. In addition, in terms of commercial context, the study will keep contributing to investigate how the use of LSS methodologies promote competitiveness through boosting customer satisfaction, simplifying the business processes and improving product quality. From the social context, the study will demonstrate how the interference of LSS approaches help business organizations establish a continuous improvement culture that stimulates worker participation and engagement. Thus, through targeting the current research lacuna, this research offers an in-depth understanding of the diverse impacts of LSS on economic, social and commercial context, delivering insightful knowledge for practical implications and future research directions.

The rest of the paper is organized as follows. The methodology and bibliometric analysis are discussed in section 2. In section 3 describes literature review and conclusions with implications are presented in the section 4 and finally, section 5 is summarized with research limitations and future research directions.

2. Methodology

2.1 Data collection

A systematic literature review (SLR) is designed based on research gaps, where the study represents an overall understanding of the LSS approach to organize the existing and available knowledge for assessing its impact on the organization. SLR is a process that primarily aims to determine, evaluate and scrutinize the fundamental studies related to a specific topic using reproductive and basic methods (Cerchione and Esposito, 2016). The SLR makes the research efforts worthwhile when compiled on trendy content for further studies (Potrich *et al.*, 2019 ; Psomas *et al.*, 2022). The following stages are structured to design this paper:

research questions clearly defined;

key information has been carefully selected from relevant sources (Scopus and Web of Science database) to provide clear and accurate insights;

keywords determination;

identifying the most appropriate articles; and

excluding data those are irrelevant after evaluation:

The current status of research field is analyzed prior to examination of the literature review (Niñerola *et al.*, 2021a). For exploring the literature from relevant sources, several keywords are determined before searching the existing literature. It is also determined the keywords based on the following research question: What is the impact of continuous improvement through LSS in business organizations? or, how does LSS extensively impact the continuous improvement in business organizations? After developing the research questions, the criteria for the inclusion of articles under the literature review process were established.

To address our research areas, we used the Scopus database to gather relevant articles pertaining to LSS. (de Freitas and Costa, 2017). Scopus is a widely accepted source for accessing articles from appropriate journals in the research domain, facilitating the collection of valuable material for systematic literature reviews and bibliometric analysis (Kipper *et al.*, 2020). The rationale for prioritizing the Scopus database in the current study over other sources (e.g., Web of Science) is that Scopus database is appropriate for in-depth reviews that involve systematic and bibliometric analyses due to its multidisciplinary nature and extensive coverage (Prakash *et al.*, 2022). Furthermore, Scopus's citation analysis strategy is

robust, enabling a more comprehensive examination of significant publications and research trends than the Web of Science database (Maia *et al.*, 2019 ; Ortega and Delgado-Quirós, 2024). Compared to Scopus database, the Web of Science database does not offer a more extensive selection of articles in relevant disciplines, such as LSS (Singh *et al.*, 2021).

In this phase, we implemented a variety of search strategies that were selected based on the most frequently occurring keywords. Keywords from a variety of keywords were typed into the search engine to locate appropriate documents for this study. As defining the search terms accurately assists to ensure capturing of relevant studies such as LSS to focus our research theme in the review process, this stage is a little bit crucial. Five search strategies were altogether used; keywords to search articles from the Scopus database were “Lean Six Sigma,” OR “LSS” OR “lean production” OR “lean manufacturing” OR “lean thinking.”

We obtained additional results from Scopus database using same search techniques, where a higher number of articles were relevant to our study. We used the five strategies with “OR” function in Scopus database to avoid duplicates, obtaining 28,972 results in total from Scopus database.

Thanks to the instruments and techniques offered by Scopus. Gathering useful papers and filtering the search keywords and terms are to be possible due to enriched databases like Scopus. For instance, 3,956 documents were found after filtering the search relevantly and among these, 2,111 that means 53.36% were articles. Moreover, we collected a large number of publications on particular topics which began from 2001 to 2024 and this continued until the 2024 which presented in the Figure 1 . In addition, after 2006, it was found that majority of the publications on the LSS topic began to flourish and from 2018, this figure continued to increase until at the end of 2023 (see Figure 1). Furthermore, we noted that we got the large number of documents which are found in English. Refining the search results, we were able to using the data collected from the chosen articles. Consequently, we filtered the searching process as follows:

Figure 1.

Evolution of publications in Scopus

Document type: articles.

Year of publication: 2001–2024 April 30.

Language: English.

Knowledge area: Business and Management.

Once the searching results (articles obtained) were filtered, we received 3956 documents from Scopus database instead of 28972 documents which were the most appropriate with our current study. It is noted that initial papers found in the first phase, those are reduced by the following a certain selection criteria (Abbate *et al.*, 2022). In the case of reviewing the literature of previous studies, we included the articles through searching keywords which are exist in the titles and abstract of the papers but which are not matched with keywords those are excluded. The excluded papers are selected after a thorough reviewing and those are irrelevant with the research questions and objectives and these are also excluded their irrelevant present systematic review. Now, after filtering we have found 3956 articles and 1994 articles are discarded due to duplications or not required for this study or irrelevant information which were not related or matched with our current study. At last, 1992 articles were selected for analysis and writing the SLR for our research (see Figure 2):

Figure 2.

Item selection process

After collecting the articles, data extraction and analysis process were carried out using bibliometric analysis in the following section.

2.2 Bibliometric analysis

As a statistical and mathematical technique, bibliometric analysis offers an exhaustive overview to illustrate the evolution in the research field based on previous scholarly studies (Abejón and Garea, 2015). Bibliometric analysis has recently gained more popularity in various disciplines (Camposeco-Negrete, 2013 ; Chen *et al.*, 2017 ; Donthu *et al.*, 2021). Additionally, bibliometric analysis is insightful for identifying recent developments and offering a schematic representation of key content by integrating of previous studies using collaborating efforts to evaluate the performance of relevant articles and journals (Campobasso and Boscia, 2023 ; Donthu *et al.*, 2021).

Bibliometric analysis is required for identifying relevant papers related to a specific field such as LSS. According to bibliometric analysis, VOSViewer software is used to create and represent the visualization of graphic maps (Van Eck and Waltman, 2010) that are chosen to perform the bibliometric analysis to define bibliometric networks in the area of LSS implementation. These networks can be synthesized for journals, researchers and publications focusing on citation, co-citation and bibliometrix package (Ding and Yang, 2022). Moreover, various types of visualization and identification co-occurrence networks are represented in the bibliometric analysis.

It is observed from the hierarchical order of the Table 1 mentioned above that Professor Antony who has published more than 150 articles on LSS and lean production while Garza-Reyes and Tortorella have published articles 48 and 34, respectively.

Table 1.

Ranking of authors based on the most publications

Order	Author	No. of publications
1	Antony J.	152
2	Garza-Reyes J.A.	48
3	Tortorella G.L.	34
4	Vinodh S.	33
5	Kumar S.	30
6	Kumar M.	29
7	Kumar V.	29
8	Devadasan S.R.	28
9	Singh D.	26
10	Chiarini A.	22

Source(s): Authors' own work

From the Table 2 above, representation of the top ten list of recognized journals over the world in the research contribution that shows a diverse range of sources focused on the production optimization and quality management. International Journal of Lean Six Sigma with 221 articles published by Emerald publisher has hold the leading position in the list, the following position placed by the International Journal of Lean Six Sigma with publications of 219 articles.

Table 2.

List of top ten journals contributing in the LSS

Order	Sources	Articles
1	<i>International Journal of Lean Six Sigma</i>	221
2	<i>International Journal of Six Sigma and Competitive Advantage</i>	219
3	<i>International Journal of Production Research</i>	165
4	<i>Production Planning and Control</i>	152
5	<i>TQM Journal</i>	149
6	<i>Total Quality Management and Business Excellence</i>	138
7	<i>Journal of Manufacturing Technology Management</i>	134
8	<i>International Journal of Productivity and Quality Management</i>	130
9	<i>International Journal of Quality and Reliability Management</i>	128
10	<i>International Journal of Productivity and Performance Management</i>	103

Source(s): Authors' own work

The bottom position placed by International Journal of Productivity and Performance Management publishing 103 articles. This journal also emphasizes how study on the LSS is broadly extended in a variety of academic domains.

The VOSviewer-generated keyword co-occurrence map shows the interconnectedness of key terms in LSS literature. The map has several noticeable nodes wherein larger nodes represents keywords that emerge more frequently. From the map, "Lean Production" is shown as the most significant node, represented by its larger size and central position in the red cluster, closely connected with other related terms such as "Agile manufacturing Systems," "Lean Manufacturing," and "Industry 4.0." in the green cluster, "Six Sigma" is the focal point. It has strong relationship with terms like "Process improvement," "TQM," and "LSS." The smaller cluster marked in the map as blue cluster which shows keywords associated with "Industrial Management," "Innovation," and "Project Management." It is clearly observed that the relationship of lean methodologies, management strategies and Six Sigma practices those are fundamental focusing points in the research of LSS domain, are emphasized by this map.

To explore the concurrence of the keywords in the research articles, we adopted the use of VOSviewer in the analysis section. In Figure 3, we can observe how many times each word shows in the research paper. This type of analysis is required to interpret the themes and contents of the documents analyzed in the research article (Rao and Shukla, 2023).

Figure 3.

Map of keyword nodes

It is visualized that three clusters are represented in Figure 3 where Red (31 items) in cluster 1, cluster 2 is represented by green (20 items) and blue (11 items) is represented by cluster 3. From the Table 3, the theme of cluster 1 is agile manufacturing systems embrace lean production and lean manufacturing principles. In the cluster 2 remains continuous improvement via LSS methodology, and cluster 3 represents innovation in industrial management boosts customer

satisfaction.

Table 3.

Clusters obtained from co-occurrence analysis

Clusters	Items	Keywords	Co-Occurrence	Themes
Cluster 1 (Red)	31	Lean production	1152	Agile manufacturing systems embrace lean production and lean manufacturing principles
		Lean manufacturing	755	
		Agile manufacturing systems	210	
Cluster 2 (Green)	20	Six sigma	1086	Continuous improvement via LSS methodology
		Lean six sigma	384	
		Continuous improvement	170	
Cluster 3 (Blue)	11	Customer satisfaction	106	Innovation in industrial management boosts customer satisfaction
		Industrial management	68	
		Innovation	54	

Source(s): Authors' own work

The co-occurrence of keywords highlights the key terms which are frequently used in the research of LSS domain. From the Table 4 , the first dominant role is placed by “lean production” with 1,152 occurrences and 3,610 of a total link strength. The second position holds by following of 1, 086 occurrences and 3,117 in terms of total link strength. Third position represents containing with 755 occurrences and a total link of strength of 1,872. It is apparent that terms such as “work simplification,” and “LSS” are widely used, highlighting their continuous improvement and significance in the process optimization. Other noteworthy terms including “sustainable development,” “process engineering,” and “agile manufacturing systems” which reflects that how efficiency and sustainability are becoming increasingly crucial in current manufacturing processes (Campobasso and Boscia, 2023).

Table 4.

Co-occurrence of keywords

Order	Keywords	Occurrences	Total link strength
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1	Lean production	1152	3610
2	Six sigma	1086	3117
3	Lean manufacturing	755	1872
4	Lean six sigma	384	1198
5	Lean	308	836
6	Work simplification	290	1773
7	Agile manufacturing systems	210	980
8	Process engineering	204	1212
9	Total quality management	179	599
9	Continuous improvement	170	598
10	Sustainable development	141	778

Source(s): Authors' own work

From the Table 5, it represents the total number of articles published in India, USA, UK, Brazil, China, Italy, Sweden, Malaysia, Spain and Australia. The ranking is arranged from the highest to lowest number of articles based on the particular country. The number of articles related to LSS published in the India (625 articles). On the other hand, the second and third position have hold by USA and UK and their number of articles are 485 and 278.

Table 5.

Country wise publications of articles

Order	Country	Articles
1	India	625
2	USA	485
3	United Kingdom	278
4	Brazil	166
5	China	131
6	Italy	116

7	Sweden	89
8	Malaysia	87
9	Spain	68
10	Australia	66

Source(s): Authors' own work

A network of global research collaborations is depicted by the country citation map wherein the USA and India are showing the central roles in international research (Figure 4). China, Spain and Brazil also display their strong connectivity focusing on their significance in the research collaborations in the field of LSS. European nations such as Spain, Germany, the Netherlands make a unique cluster reflecting strong regional collaboration in the research field. In addition, In the line of Asian regions, Malaysia, Taiwan and Pakistan are inter-connected and they contribute to the international research with smaller nodes in the map.

Figure 4.

Country citation map

After completion of the bibliometric analysis, a literature review is required for justification and defining the research area by reviewing previous studies systematically. However, in the next section, summarization key findings from previous literature is discussed with references in the Table 6 .

Table 6.

Recapping table with the references analyzed in each section

Areas	References
4.1 Conceptualization of LSS	Antony (2020) ; Antony and Chakraborty (2024) ; Asif (2021) ; Costa <i>et al.</i> (2021) ; Nascimento <i>et al.</i> (2020) ; Null <i>et al.</i> (2020) ; Sunder and Antony (2020)
4.2 Agile manufacturing systems embrace lean production, lean manufacturing principles	Naylor <i>et al.</i> (1999) ; Buer <i>et al.</i> (2018) ; Cao <i>et al.</i> (2024) ; Ding <i>et al.</i> (2023) ; Fullerton <i>et al.</i> (2014) ; Gunasekaran <i>et al.</i> (2019) ; Jastia and Kodali (2015); Kamble <i>et al.</i> (2020) ; R. Panizzolo <i>et al.</i> (2012) ; Rosin <i>et al.</i> (2020) ; R. Sundar <i>et al.</i> (2014) ; Y. Y. Yusuf <i>et al.</i> (1999)
4.3 Continuous improvement via LSS methodology	J. Antony <i>et al.</i> (2017) ; Indrawati and Ridwansyah (2015) ; Morgan (2019a , 2019b) ; Narottam <i>et al.</i> (2019) ; Pepper and Spedding (2010) ; Salah <i>et al.</i> (2010) ; Timans <i>et al.</i> (2016)
4.4 Innovation in industrial management boosting customer satisfaction	Alcaide-Muñoz and Gutierrez-Gutierrez (2017) ; Alexander <i>et al.</i> (2022) ; Arcidiacono and Pieroni (2018) ; Irani <i>et al.</i> (2004)

Source(s): Authors' own work

3. Literature review

3.1 Conceptualization of lean six sigma

The LSS concept was first introduced and integrated into the USA in 1986 by the George Group and the LSS term was

included in teaching in 2003 as part of Six Sigma (Albliwi *et al.*, 2015 ; Antony and Chakraborty, 2024 ; Asif, 2021 ; Chakravorty and Shah, 2012 ; Timans *et al.*, 2012 ; Vinodh *et al.*, 2014). Lean Six Sigma concept is described as reducing the defects or variations of produced goods and products and services and eliminating the manufactured wastage to obtain efficiency and achieve organizational goals and objectives (Antony, 2020 ; Raval and Kant, 2017). It helps to achieve the organization's quality, efficiency and customer satisfaction (Raja Sreedharan and Raju, 2016). Moreover, Lean Six Sigma uses a variety of tools and techniques such as statistical methods, process mapping, cause and effect diagrams and others through considering main purposes like waste reduction and minimizing defects or variations (Costa *et al.*, 2021 ; Thomas *et al.*, 2016). If we deeply classify the lean Six Sigma, we can get two key terms which are combined to integrate including lean manufacturing and Six Sigma. Lean means identifying the value to satisfy the customer through delivering quality products and services (Tengtarto *et al.*, 2022). The six-sigma term is defined as control processes in which the production variations reduction by minimizing the defective issues during production (Patel and Desai, 2018). Six Sigma follows the DMAIC principles in which defining means identifying the problems and measuring refers to the measurement of current activities and collecting data based on that performance and analyzing means finding out the root causes behind the problems, deficiencies and improvement refers to the development and implementation of solutions to improve the processes during manufacturing and finally, controlling means the sustaining the improvement and preventing the re-occurrence to the defective issues or problems in the manufacturing (Nandakumar *et al.*, 2020). Thus, lean and Six Sigma have been integrated into Lean Six Sigma which helps to go forward with continuous improvement within the manufacturing systems and organizational activities (Null *et al.*, 2020). Lean Six Sigma is employed and maintained to achieve efficiency and quality through improving the processes reducing the defects from the manufactured products and generating services offered by the organization (Nascimento *et al.*, 2020). It is demanded that for successful implementation and achieving operational and environmental benefits, Lean Six Sigma should be integrated with sustainability tools and techniques (Cherrafi *et al.*, 2017 ; Kaswan and Rathi, 2020 ; Ruben *et al.*, 2018).

3.2 Agile manufacturing systems embracing lean production and lean manufacturing principles

Agile manufacturing systems strive to reduce production cost, minimize waste and improve efficiency by integrating lean production and lean manufacturing principles (Cao *et al.*, 2024 ; Gunasekaran *et al.*, 2019 ; Naylor *et al.*, 1999). Lean production focuses on streamline processes, eliminating pointless procedures and continual improvement, which support the agile manufacturing systems' demands for rapid customization in the organizations (Malviya, 2021). This integration allows the organizations to keep customers satisfied by providing high-quality products and remain competitive in the constantly shifting marketplaces (Ahmed and Huma, 2021 ; Mathiyazhagan *et al.*, 2021).

In contrast, Incorporation the principles of lean manufacturing into agile manufacturing systems offers a well-rounded approach by combining on the efficiency and flexibility (Ding *et al.*, 2023). Lean manufacturing makes ensure that these adjustments are made possible using minimum amount of resources while agile manufacturing systems helps the organizations to respond quickly to the new trends (Agarwal *et al.*, 2023 ; Dubey and Gunasekaran, 2015). The togetherness between two approaches generates an environment in the manufacturing systems which enables to be responsive, economical and sustainable in the landscapes of the marketplace (Touriki *et al.*, 2021). Thus, in a highly competing international marketplace, the fusion of lean and agile principles allows the companies in achieving greater innovation, faster production time and better operational performance during manufacturing processes (Nieuwenhuis and Katsifou, 2015). Now, we have attempted to represent about how agile manufacturing systems embrace lean production and lean manufacturing principles as follows:

Customer Focus: Agile and lean manufacturing systems focus on a strong emphasis to identify, understand and respond swiftly to meet the customer needs (Oliveira-Dias *et al.*, 2021). Agile manufacturing provides the opportunity of flexibility and responding to the customers' needs and preferences quickly. On the other hand, lean manufacturing's main purpose to remove the waste and improving the processes of the production which ultimately leads to customer value.

Adaptability and Flexibility: Continuous improvement and JIT manufacturing principles allow to provide flexibility and facilitate the adaptability to the market changes. Moreover, it also provides for adjustment to new processes and it carries

out to further improvement of manufacturing processes with the responding of market challenges and problems (Iyengar and Bharathi, 2018).

Cross-functional Collaboration: Agile manufacturing provides the collaboration and cooperation among different functions, teams and departments for facilitating the immediate decision-making and quickly solving the problems occurred in the manufacturing process (Brandl *et al.*, 2021). While lean manufacturing principles promotes the cross-departmental and functional group work to find out and reduce the wastage from the processes, improving processes and ultimately it leads to the overall performance (De Oliveira *et al.*, 2019).

Continuous Improvement: Continuous enhancements of processes like practices of changing and trendy cultures and implementation of innovative systems and adopting adaptability over the time, these are driven by agile manufacturing (Ito *et al.*, 2023). While lean principles such as small improvement of the production processes, kaizen and incremental changing over the time and situation to ensure the efficiency of the organization (De Oliveira *et al.*, 2019).

Overall, the agile manufacturing embraces the lean production and lean manufacturing principles to improve the processes and allow the adaptability and flexibility and responding to the customers' needs and demands.

3.3 Continuous improvement via lean six sigma methodology

LSS is an integrative, powerful methodology that blends the lean and six sigma approaches to foster continuous development within organizations (Gaikwad and Sunnapwar, 2020 ; Narottam *et al.*, 2019 ; Patel and Patel, 2021). It generates cutting-edge opportunities for efficiency and promotes the long-term success by continuously improving business processes (Pepper and Spedding, 2010 ; Skalli *et al.*, 2024). Lean approach minimizes waste and streamlines business operations to improve efficiency, while six sigma reduces defects and variations in manufacturing processes to ensure quality through statistical analysis (Daniyan *et al.*, 2022 ; Ruben *et al.*, 2018). By identifying deficits and using precise measurements, LSS enables organizations to execute improvements through lowering costs, achieving operational excellence, promoting customer satisfaction (Arcidiacono *et al.*, 2016 ; Bilgin Turna, 2023). Consequently, these initiatives improve ultimately total probability and financial stability within organizations. So, it is apparent from the reviewing of previous studies, LSS approach is developed following a methodical, step-by-step process (Wang *et al.*, 2020). The very first step, to pinpoint the problems and errors and draw attention that need improvement (Oliver *et al.*, 2019). Next, organizations assess their current performance for figuring out what types of improvements required to remove deviations (Hardy *et al.*, 2021). In the third step, the data are examined to determine the root causes of deficiencies within organizational processes (Swarnakar *et al.*, 2020).[AQ] Following that, Potential solutions are developed and new techniques, systems and tools are placed into business processes where required improvement (Uluskan, 2019). Last but not least, by continual monitoring and feedback, this methodology highlights controlling to achieve sustainability and enhance improvements over time (Erdil *et al.*, 2018). Thus, LSS as a systematic approach drives continuous improvement and fosters long-term success.

3.4 Innovation in industrial management boosting customer satisfaction

Innovation through LSS has transformed the industrial management by promoting continuous improvement, optimizing business processes, which ultimately boost customer satisfaction (Goshime *et al.*, 2019 ; Rahardjo *et al.*, 2024). Based on previous studies, in industrial management, LSS emphasizes on continual development, data-driven decision-making and cross-functional collaboration (Belhadi *et al.*, 2023 ; Tissir *et al.*, 2024). By infusing innovation with systematic, data-driven approach of the LSS, organizations can minimize waste, eliminate inefficiencies and deploy innovative solutions that substantially increase their operational performance (Clancy *et al.*, 2023 ; Rajić *et al.*, 2023). This powerful integration creates an environment by maintaining procedures that are adaptable and sensitive to changing market demands (Alcaide-Muñoz and Gutierrez-Gutierrez, 2017 ; Antony *et al.*, 2016). As industries emerge, LSS enables organizations to satisfy consumer expectations by providing superior quality and faster turnarounds in businesses, resulting in enhanced overall productivity (Raval *et al.*, 2020 ; Sunder *et al.*, 2019). However, innovation-driven continual development raises customer satisfaction by continually filtering goods and services to meet the demands of customers (Sunder *et al.*, 2020). In addition,

In the line with the use of LSS, organizations can enhance value by reducing lead times, improving product quality, realizing reduced costs which indirectly benefit customers (Kumar and Reinartz, 2016). As the very structured methodology, successful implementation of LSS ensures efficiency and quality by adopting innovation ideas in industrial management (Antony *et al.*, 2016 ; Sony *et al.*, 2020). So, constant innovation creates improved brand reputation, customer loyalty and long-term success thereby keeping organizations competitive, forward-thinking and responsive to their customers swiftly (Gifford and Newmeyer, 2019 ; Khan *et al.*, 2022).

4. Key barriers in LSS implementation

Several significant challenges emerge in LSS implementation, although it has enormous potential to enhance operational excellence (Lameijer *et al.*, 2021 ; Tissir *et al.*, 2024). A key barrier in using LSS methodologies is the substantial investment of efforts, time, training and resources needed for successful implementation (Vinodh and Shimray, 2023). Numerous initiatives fail due to limited resources, insufficient preparation and absence of strong dedication, which hinders organizations to fully embrace them (Sithambaram *et al.*, 2021). These obstacles degrade trust in LSS and trigger questions regarding its practicality and efficiency.

4.1 Barriers to integration between LSS methodologies and digital tools

A dearth of integration between LSS methodologies and digital tools is major obstacle (Laux *et al.*, 2017). Organizations often face challenges in integrating digital tools with LSS principles, despite the potential of digital tools to improve processes (Citybabu and Yamini, 2023). Their inability to overcome this gap hinders their progress toward operational growth (Johansson *et al.*, 2024). So, the collaboration between LSS and digital transformation may synergize to close this gap, fostering innovation in problem-solving and enhancing efficiency (Capolupo *et al.*, 2024).

4.2 Leadership and cultural barriers in LSS implementation

The successful implementation of LSS is also impeded by leadership challenges and cultural barriers (Marolla *et al.*, 2022). Because, For numerous organizations, leaders who either failure to foster a culture of continuous improvement or are incapable of comprehending the implementation of LSS create a substantial source of concern (Marolla *et al.*, 2022). Absent a robust determination to leadership change, LSS initiatives frequently fail. So, to accomplish long-term success in organizations and advance LSS maturity, it is essential to transcend leadership and cultural barriers (Laureani and Antony, 2019 ; Marolla *et al.*, 2022).

5. Conclusions and implications

5.1 Conclusions

Informed by the insights derived from the bibliographic and bibliometric analysis of this study, the conclusions have been drawn. The adoption of the LSS principles results in the continuous and transformative development of organizations. The adoption of the LSS principles results in the continuous and transformative development of organizations. The findings of this study highlight not only superior quality but also emphasize enhanced employee engagement and notable cost savings. This robust methodology fosters a culture of accountability and creativity, enabling each member to enhance productivity and achieve long-term objectives. The LSS journey fosters a mind-set of unwavering quest for excellence. The framework provided by LSS enables organizations to overcome challenges as they innovate and adapt, thereby guaranteeing a competitive and sustainable position in a changing market. By incorporating LSS principles, organizations can enhance their capacity to deliver consistent value, maintain their competitive advantage and optimize resources. Finally, not only is LSS a strategic force that encourages organizations toward sustainable profitability, but it also serves as a tool for continuous improvement.

5.2 Theoretical implications

The research findings provide valuable theoretical insights that contribute to the understanding of the application of LSS to the continual improvement of business processes. The current study outlines five significant theoretical implications. First, the fusion of Six Sigma and Lean principles promotes a unified strategy that enhances quality and process efficiency via elimination and maximization of value. Second, the study underscores the benefits of LSS's data-driven decision-making, which promotes continuous improvement by enabling more informed and effective interventions that facilitate problem-solving. Third, it highlights the transformative influence of a culture of perpetual development, in which all levels of an organization are committed and empowered to leading continuous innovations and advancements. Fourth, LSS effectively

combines operational strategies with customer demands, markedly enhancing customer satisfaction and promoting maintaining customer loyalty. Finally, the study reveals Continuous performance improvement is the driving force behind long-term organizational success, and LSS is not only a methodology but also a strategic force that provides a competitive advantage in the marketplaces. Thus, this theoretical foundation emphasizes the transformative capacity of LSS in the business environment, thereby supporting both strategic growth and operational excellence.

5.3 Practical implications

The study provides a number of practical implications that are both insightful and have the potential to enhance the comprehension of continuous improvement in business processes through LSS. First, it identifies the necessity of implementing LSS tools in a variety of industries beyond manufacturing to boost customer satisfaction, improve quality and accomplish operational excellence. Second, this research points out that organizations must personalize LSS principles to align with their own regulatory structures and organizational cultures to achieve optimal outcomes. Third, by identifying and eliminating waste, reducing variability and simplifying business processes, the total productivity is significantly enhanced through the successful implementation of LSS. Fourth, Sustainable development is promoted by LSS through the enhancement of revenues and the reduction of operating costs, which in consequently enhances economic performance and promotes customer satisfaction. Fifth, Standardizing procedures by eliminating non-value-adding activities, such as redundant paperwork, will accelerate operations and improve resource efficiency. Finally, Finally, LSS promotes empowerment and promotes an environment of innovation by fostering continual development, collaboration and creatively problem-solving, which optimizes the use of organizational resources and push to organizational success. Thus, these practical ramifications provide a useful and realistic road-map organizations seeking to sustainable success through LSS.

6. Research limitations and future research directions

6.1 Research limitations

Despite study’s valuable insights but it has several shortcomings as well. First, the study concentrates solely on manufacturing sectors, which creates a gap in understanding how their results are applicable to other sectors such as education and health care. Second, the potential of LSS for unleashing greater innovation is restricted by its dearth of fusion with emerging technologies like automation and artificial intelligence. Third, numerous analyses rely heavily on qualitative approaches, which indicates the lack of validation of findings with data-driven insights through application of quantitative studies. Finally, this research cannot provide a robust and united model for achieving sustainability goals by integrating LSS, particularly in key industries such as construction and automobiles.

6.2 Future research directions

Further research can be suggested for this study by using longitudinal investigation method to overcome these limitations to explore how LSS can be implemented in sectors like health care and educational institutions, where unique opportunities and challenges may evolve. Moreover, the unlocking of the potentiality of driving efficiency and innovation through integrating LSS tools and techniques such as artificial intelligence and automation. Additionally, future research can be explored by applications of quantitative methods to ensure the data-driven insights by checking the validity of this study’s findings. Finally, building a strong and cohesive model that links LSS with sustainable development targets across industries like construction and automobiles, which will be significant for establishing more broadly, long-term impacts on organizations.

However, According to Centobelli *et al.* (2020) , the clusters that were specified for classifications of the bibliometric analysis and the articles selected are divided into primary three clusters presented the Table 7 :

Table 7.

Main topics and future research avenues

Cluster	Exemplary references	Future research directions
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Red (cluster 1): research on agile manufacturing systems embrace lean production and lean manufacturing principles	Jasti and Kodali (2015) ; Kamble <i>et al.</i> (2020) ; Roberto Panizzolo <i>et al.</i> (2012) ; Rosin <i>et al.</i> (2020) R Sundar <i>et al.</i> (2014) ; Yahaya Y Yusuf <i>et al.</i> (1999)	To investigate the impact of soft lean manufacturing and industry 4.0 To explore the effects of environmental factors on the integration of lean manufacturing and industry 4.0 To ascertain lean performance, lean sustainability, kanban areas beyond lean production and lean manufacturing principles To propose the research frameworks/models which can create a revolution in the field of lean manufacturing or production To analyze new applications or utilization of industry 4.0 technologies integrated with lean principles
Green (cluster 2): continuous improvement via LSS methodology	(Jiju Antony <i>et al.</i>, 2017 ; Indrawati and Ridwansyah, 2015 ; Narottam <i>et al.</i>, 2019 ; Pepper and Spedding, 2010) ; Salah <i>et al.</i> (2010)	To strengthen the improvement of six sigma by integrating internet and information technology To investigate quality based analysis on value and non-value activities To look into the integration of LSS and supply chain for ensuring sustainability To explore the suitability of LSS under the theoretical support of total quality management

Blue (cluster 3): Innovation in industrial management boosting customer satisfaction	Alcaide-Muñoz and Gutierrez-Gutierrez (2017) ; Alexander <i>et al.</i> (2022) ; Arcidiacono and Pieroni (2018) ; Zahir Irani <i>et al.</i> (2004)	To explore the relationship between six sigma and ambidexterity through testing proposed theories and frameworks To analyze the factors of failure of six sigma due to both a combination of exploitation and exploration orientations To find out the successful competitive position of manufacturing organizations due to use of ambidexterity integrated with six sigma methodologies To conduct further studies in the SME manufacturing companies of LSS through using interviews and online surveys
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Source(s): Authors' own work

agile manufacturing systems embrace Lean Production and Lean Manufacturing principles;

continuous improvement via Lean Six Sigma methodology; and

innovation in industrial management boosts customer satisfaction.

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DETAILS

Business indexing term:	Subject: Product quality Quality management Total quality Six Sigma Lean manufacturing
Subject:	Author productivity; Product quality; Bibliometrics; Literature reviews; Quality management; Total quality; Systematic review; Six Sigma; Sustainability; Efficiency; Lean manufacturing; Trends; Keywords
Identifier / keyword:	Bibliometric Analysis; Lean Six Sigma; Continuous Improvement; Systematic Literature Review
Publication title:	International Journal of Lean Six Sigma; Bingley
Volume:	16
Issue:	5
First page:	1252
End page:	1275
Publication year:	2025
Publication date:	2025
Publisher:	Emerald Group Publishing Limited
Place of publication:	Bingley
Country of publication:	United Kingdom

Publication subject:	Business And Economics--Management
ISSN:	20404166
e-ISSN:	20404174
Source type:	Scholarly Journal
Language of publication:	English
Document type:	Review
Publication history :	
Online publication date:	2025-03-14
Milestone dates:	2024-08-07 (Received); 2024-10-18 (Rev-Recd); 2025-02-12 (Accepted)
Publication history :	
First posting date:	14 Mar 2025
DOI:	https://doi.org/10.1108/IJLSS-08-2024-0173
ProQuest document ID:	3283336470
Document URL:	https://www.proquest.com/scholarly-journals/continuous-improvement-through-lean-six-sigma/docview/3283336470/se-2?accountid=10016
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Last updated:	2026-02-12
Database:	ProQuest One Academic

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